

University of Al-Kafeel

College of Dentistry



LEC-5

Adrenergic agonist Drugs

Asst.Prof.Dr.Yasmeen Ali Hussein

Learning Objectives



By the end of this lecture, you should be able to

- Explain the mechanism of action of adrenergic agonists .
- Identify the therapeutic uses and clinical indications of adrenergic agonists
- Recognize the main adverse effects, contraindications, and precautions associated with adrenergic agonist drugs.

Sympathomimetics

Are drugs that produce effects similar to sympathetic nerve stimulation.

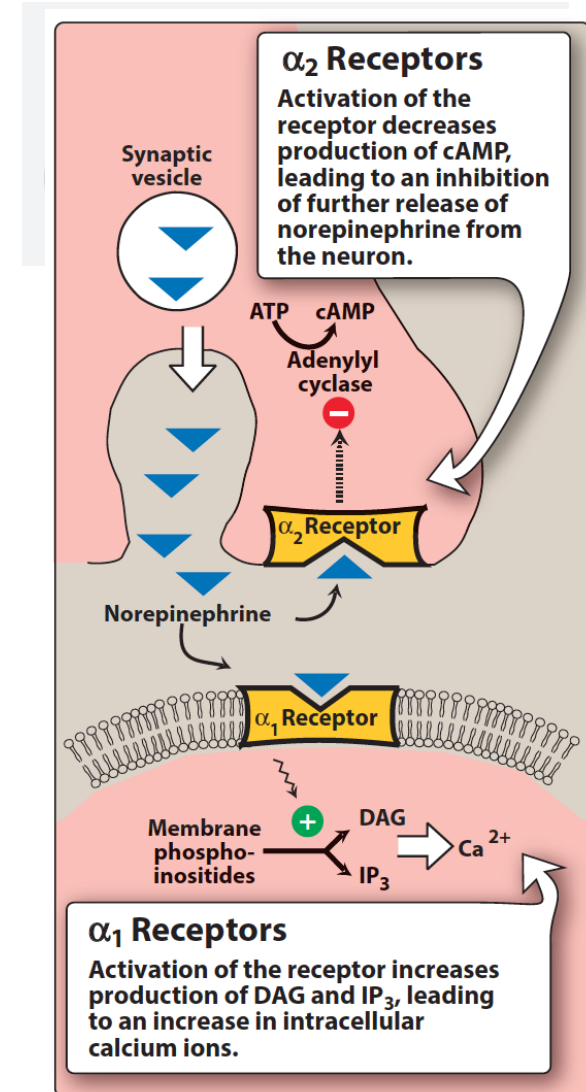
❑ Adrenergic agonist

❑ Classification of adrenoceptors (Adrenergic receptors)

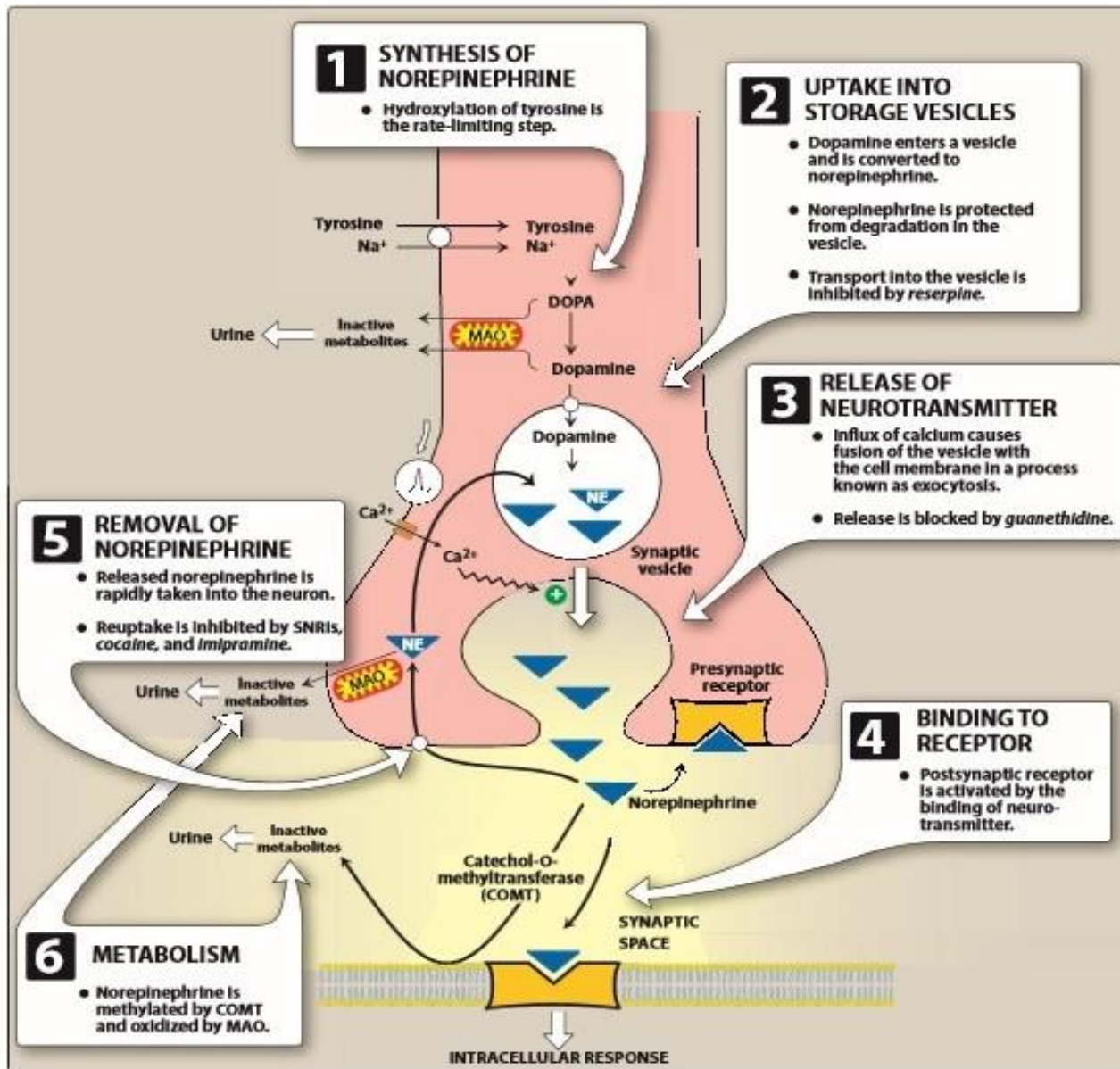
- two main α -receptor subtypes, α_1 and α_2 , each divided into three further subtypes
- three β -adrenoceptor subtypes (β_1 , β_2 , β_3)

• Second messengers:

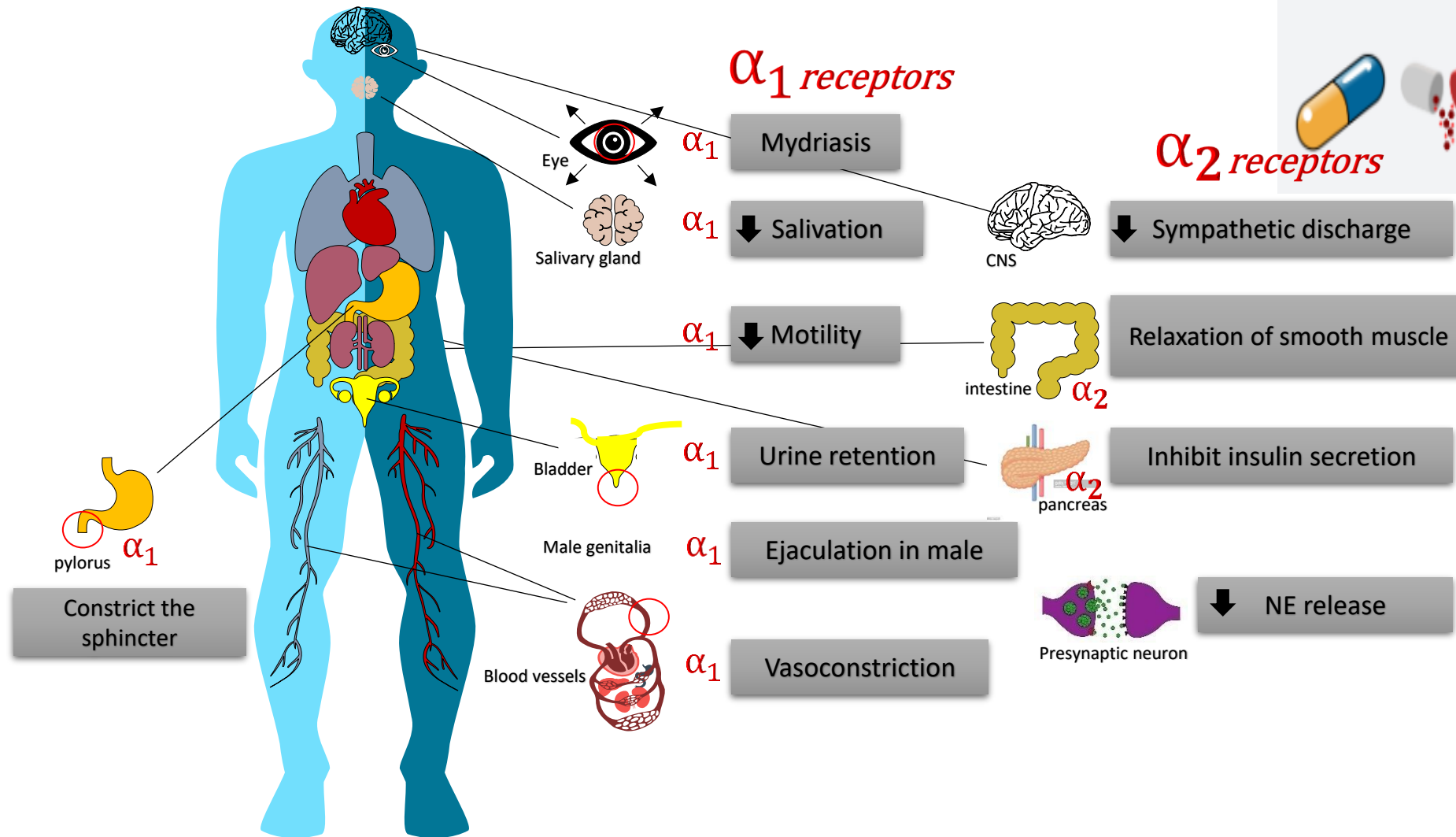
- α_1 -receptors activate phospholipase C, producing inositol trisphosphate and diacylglycerol as second messengers
- α_2 -receptors inhibit adenylate cyclase, decreasing cAMP formation
- all types of β -receptor stimulate adenylate cyclase.



Synthesis of Norepinephrine

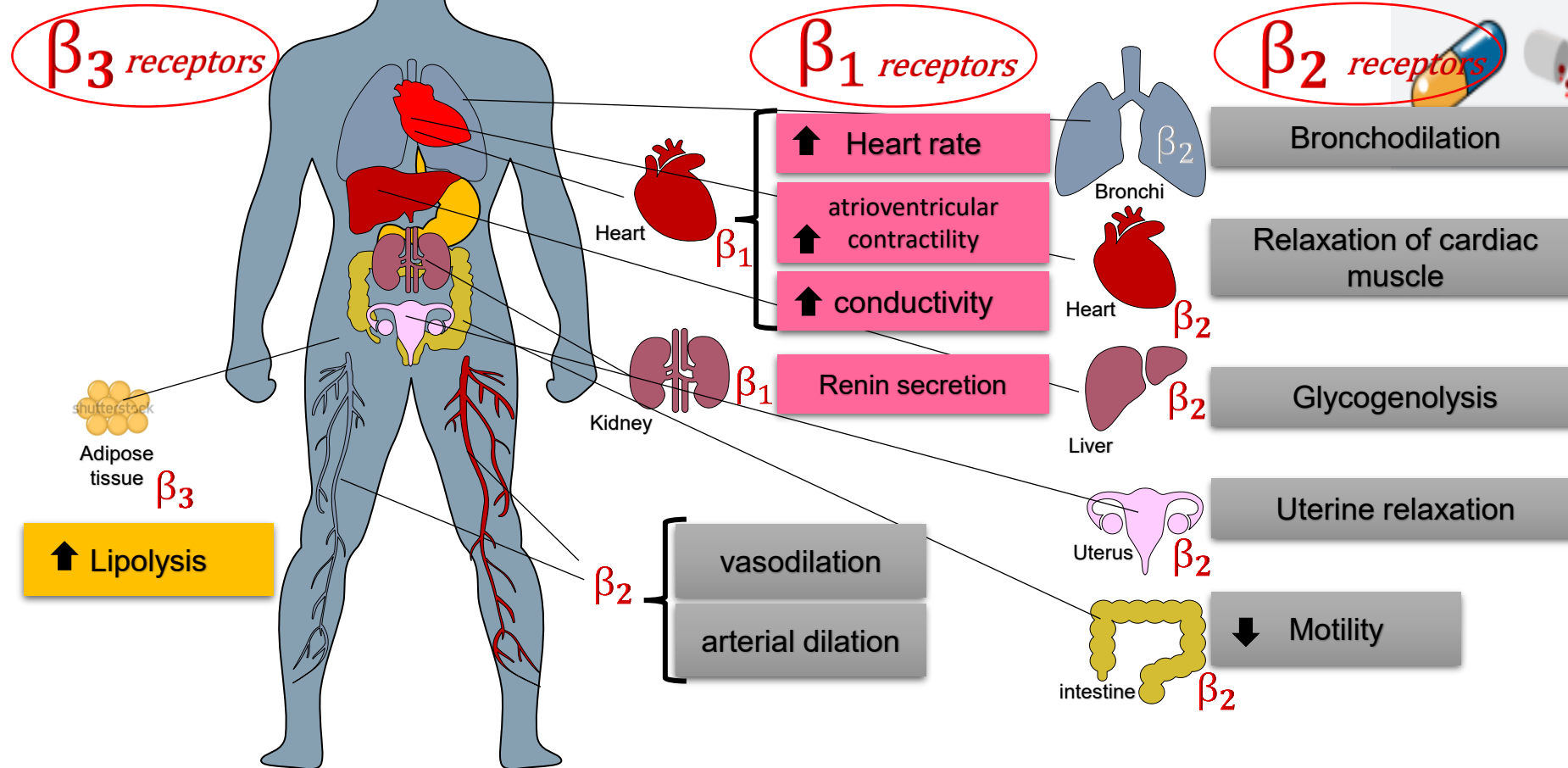


Distribution of α Receptors in the body



Distribution of B Receptors in the body

Beta receptors



ADRENERGIC AGONISTS Classification



1. according to chemical structure:

(Catecholamines & Non-catecholamines)

- **Catecholamine :**

- Catecholamines are compounds containing a catechol moiety (a benzene ring with two adjacent hydroxyl groups) and an amine side-chain. Pharmacologically, the most important ones are:

- **Noradrenaline**
- **Adrenaline**
- **Dopamine**
- **Isoproterenol** (isoprenaline), a synthetic drug
- **Dobutamine**, synthetic drug, act as selective agonist for β_1 receptors

• 2. according to receptor selectivity:

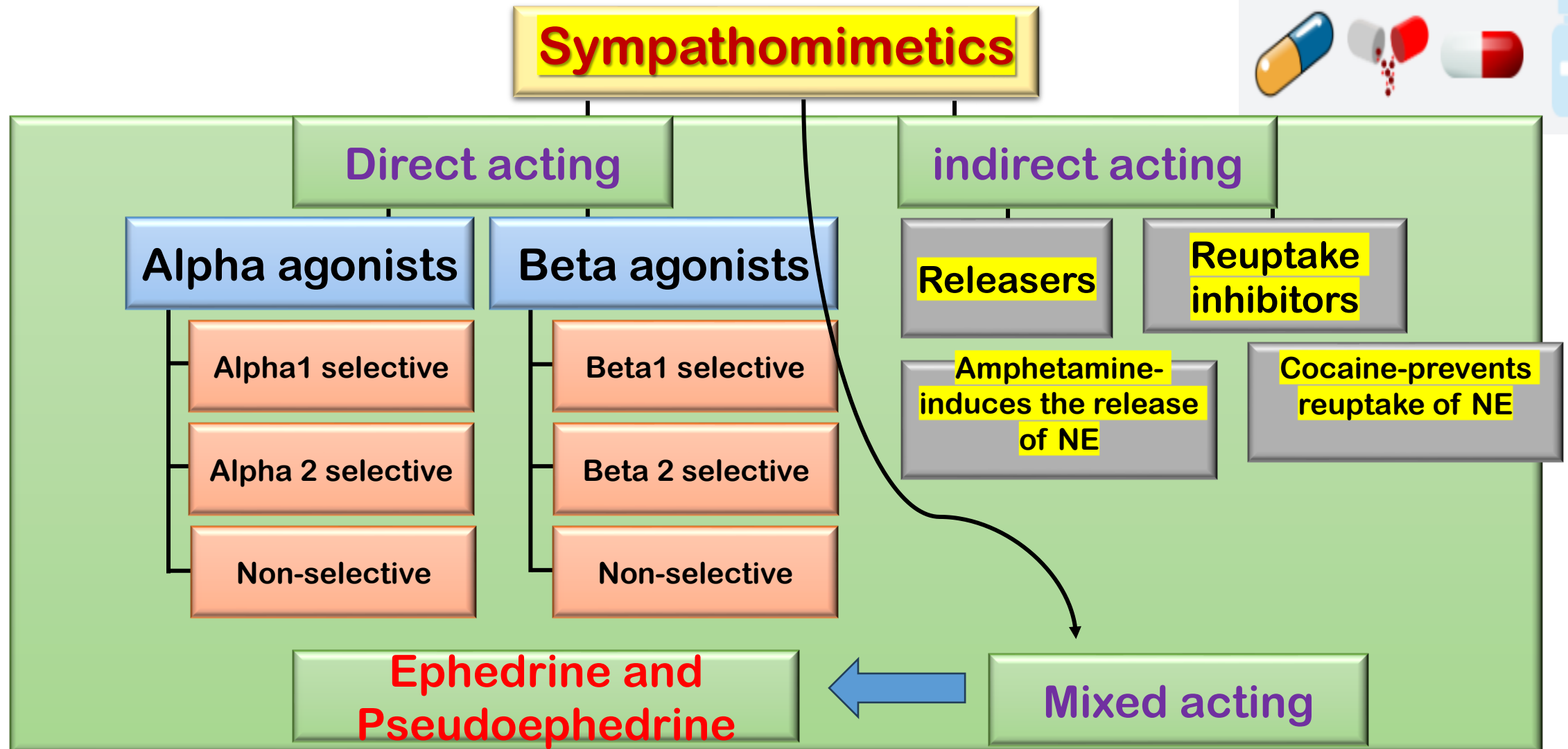
- *Selective α_1 agonists*
- ***Selective α_2 agonists***
- *Selective β_1 agonists*
- ***Selective β_2 agonists***

ADRENERGIC AGONISTS Classification

3. according to the mode of action:

- **Direct-acting agonists:** These drugs act directly on α or β receptors. Examples include *adrenaline*, *noradrenaline*, *isoproterenol*, *phenylephrine*etc.
- **Indirect-acting agonists:** These agents may block the uptake of noradrenaline (**uptake inhibitors**) e.g.: *cocaine* or are taken up into the presynaptic neuron and cause the **release of noradrenaline** e.g. *amphetamines*.
- **Mixed-action agonists:** Some agonists, such as *ephedrine* and *pseudoephedrine* have the capacity both to stimulate adrenoceptors directly and to release noradrenaline from the noradrenergic neuron

Sympathomimetics classifications



Sympathomimetics classifications



➤ **Direct-acting:** (α agonist, β agonist, α and β agonist)

▪ **Catecholamines**

- Naturally occurring- Norepinephrine (NE), Epinephrine (E), dopamine
- Synthetic- isoproterenol and dobutamine

▪ **Non catecholamines** : e.g. Oxymetazoline

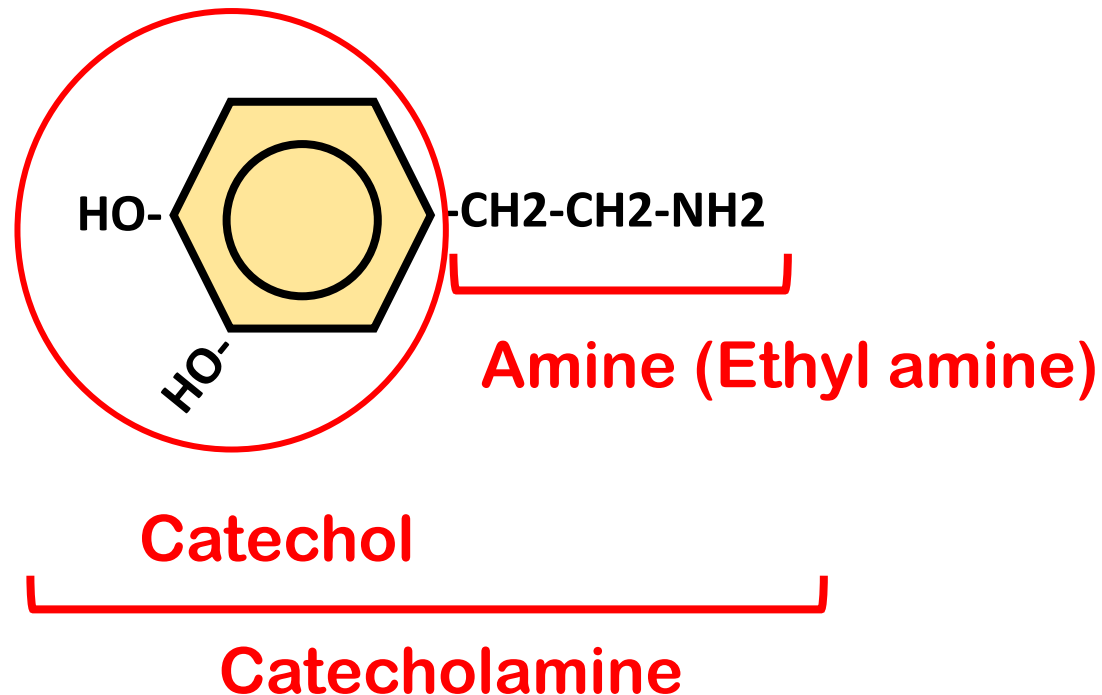
➤ **Indirect-acting:**

- Amphetamine- induces the release of NE
- Cocaine-prevents reuptake of NE
- **Mixed-acting:** Ephedrine and Pseudoephedrine

A-Direct-acting adrenergic receptor CATECHOLAMINES

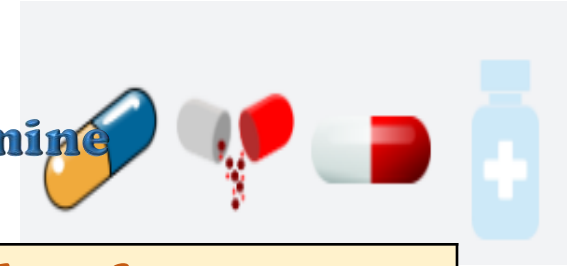


- Are compounds containing a catechol moiety (a benzene ring with two adjacent hydroxyl groups) and an amine side-chain .



A-Direct-acting adrenergic receptor

Differences between Catecholamine & non-Catecholamine



Catecholamines	Non-catecholamines
High potency	Low potency
Do not readily penetrate into CNS	Readily penetrate into CNS
Metabolized by MAO & COMT	Not metabolized by MAO & COMT
Not given orally (except isoprenaline)	given orally
Short duration of action	Long duration of action
Less lipid soluble	High lipid solubility
e.g. adrenaline, dopamine, isoprenaline	e.g. ephedrine, phenylephrine, amphetamine

A-Direct-acting adrenergic receptor CATECHOLAMINES



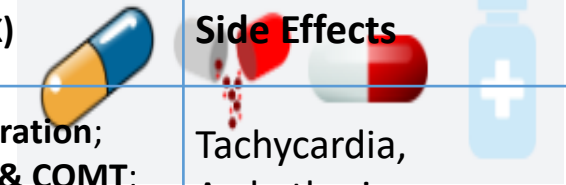
Natural

- **Norepinephrine (Noradrenaline)**, a transmitter released by sympathetic nerve terminals
- **Epinephrine (Adrenaline)** a hormone secreted by the adrenal medulla

Synthetic

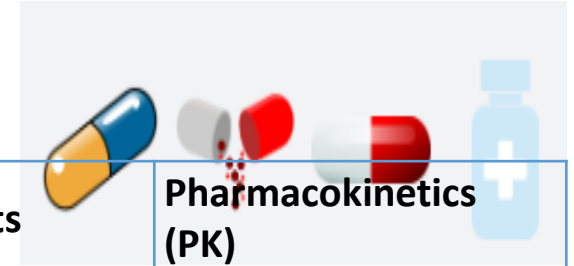
- **Dopamine**, the metabolic precursor of norepinephrine epinephrine
- **Isoproterenol (previously isoprenaline)**, a synthetic derivative of norepinephrine, is not present in the body
- **Dobutamine**, a synthetic drug, acts as a selective agonist for β_1 receptors

Catecholamines – Direct-Acting(Non-Selective)



Drug	Main Receptors	Actions	Therapeutic Uses	Pharmacokinetics (PK)	Side Effects
Epinephrine (Adrenaline)	$\alpha_1, \alpha_2, \beta_1, \beta_2, \beta_3$	$\uparrow$ BP (α_1), $\uparrow$ HR (β_1), bronchodilation (β_2), $\uparrow$ glucose/lipolysis(β_3)	Anaphylaxis (DOC), Cardiac arrest, Bronchospasm, Local bleeding, Adjunct in local anesthesia	Rapid onset, short duration; metabolized by MAO & COMT ; not effective orally ; given IV, IM, SC, inhalation ; metabolites: VMA, metanephrine	Tachycardia, Arrhythmia, Hypertension, Hyperglycemia
Norepinephrine (Noradrenaline)	$\alpha_1 > \alpha_2 > \beta_1$	Vasoconstriction $\rightarrow$ $\uparrow$ BP (systolic & diastolic); Reflex $\downarrow$ HR	Shock refractory to other drugs	Rapidly inactivated by MAO & COMT ; IV only due to short half-life.	Tissue ischemia, bradycardia
Isoproterenol	$\beta_1 = \beta_2$	$\uparrow$ HR/contractility, vasodilation $\rightarrow$ $\downarrow$ TPR, bronchodilation	AV block, cardiac arrest (rarely used for asthma)	Rapid hepatic metabolism by COMT , short duration; IV, inhaled	Tachycardia, arrhythmia, hypotension
Dopamine	D_1, β_1, α_1 (dose-dependent)	Low dose $\rightarrow$ renal vasodilation; Mid dose $\rightarrow$ $\uparrow$ CO; High dose $\rightarrow$ vasoconstriction	Shock with renal shutdown	Very short half-life , given IV infusion ; metabolized by MAO-B & COMT	Nausea, arrhythmia, hypertension
Dobutamine	$\beta_1 > \beta_2$	$\uparrow$ CO without $\uparrow$ HR much	Acute heart failure, cardiogenic shock, post-surgery support	Half-life ~2 min, IV infusion only , rapid hepatic metabolism	Arrhythmias, tolerance with prolonged use

α -Adrenergic Agonists



Drug	Selectivity	Actions	Therapeutic Uses	Adverse Effects	Pharmacokinetics (PK)
Phenylephrine	α_1	Vasoconstriction, $\uparrow$ BP, reflex bradycardia	Nasal decongestant, SVT, mydriasis	Hypertension, rebound congestion	Not metabolized by COMT , longer action than catecholamines; oral, IV, topical forms
Xylometazoline / Oxymetazoline	α_1	Nasal/ocular vasoconstriction	Nasal decongestant, eye redness relief	Rebound congestion, insomnia	Topical nasal/ocular; systemic absorption possible causing CNS effects
Methoxamine	$\alpha_1 + \alpha_2$	Vasoconstriction	Treat intraoperative hypotension, SVT	Excess vasoconstriction	Parenteral; longer duration than catecholamines
Clonidine	α_2 central & peripheral	$\downarrow$ Sympathetic outflow, $\downarrow$ NE release	Hypertension, opioid withdrawal, migraine, menopausal flushing	Sedation, rebound HTN	Well absorbed orally , long duration; crosses BBB
Methyldopa	Converted to methylnorepinephrine (α_2)	"False transmitter," $\downarrow$ NE release	Hypertension in pregnancy	Fatigue, hepatotoxicity, anemia	Crosses BBB; prodrug; metabolized in neurons to active form

β -Adrenergic Agonists



Drug	Receptors	Actions	Uses	Side Effects
Salbutamol (Albuterol)	β_2	Bronchodilation, uterine relaxation, $\uparrow K^+$ uptake	Asthma, premature labor, hyperkalemia	Tremor, tachycardia, hypokalemia
Terbutaline	β_2	Bronchodilation (similar to Salbutamol)	Asthma (PO or inhaled)	Same as above
Salmeterol / Formoterol	β_2	Long-acting bronchodilation (12 h)	Asthma control + inhaled steroids (not acute)	Tachycardia, tremor
Dobutamine	β_1	$\uparrow$ Contractility, $\uparrow$ CO	Heart failure, shock	Arrhythmia

Indirect-Acting Sympathomimetics



Drug	Mechanism	CNS / Peripheral Effects	Uses	Adverse / Toxic Effects
Amphetamine	Releases NE + blocks reuptake	↑Alertness, ↓Fatigue, ↓Appetite; ↑BP & HR	Narcolepsy, ADHD, appetite suppression	HTN, insomnia, psychosis, dependence
Cocaine	Inhibits NE reuptake	↑BP, vasoconstriction, euphoria	Local anesthetic (rare) / Drug abuse	Seizures, dependence, HTN, MI
Tyramine	Displaces stored NE	↑BP if MAO is inhibited → hypertensive crisis	— (found in cheese, wine etc.)	Dangerous with MAO inhibitors

Mixed-Action Adrenergic Agonists



- **Ephedrine and Pseudoephedrine**

- Induce the release of NA and activate adrenergic receptors

- Release stored NA from nerve endings and directly stimulate both α and β receptors. (non-selective) Thus, they produce a wide variety of adrenergic actions.

Not metabolized by COMT and MAO; thus, these drugs have a long duration of action.

They have excellent absorption orally and penetrate into the CNS; however, *pseudoephedrine has fewer CNS effects.*

Therapeutic uses of ephedrine:

- Chronic treatment asthma
- Cold-preparations
- Nasal decongestant (systemic use)
- To raise blood pressure.

S.E.: Hypertension, tachycardia, insomnia, dependence.

Summary Clinical Correlations



Condition	Drug of Choice	Mechanism / Rationale
Anaphylaxis	Epinephrine	α vasoconstriction + β_2 bronchodilation
Cardiac Arrest	Epinephrine	β_1 inotropy/chronotropy
Shock (with renal failure)	Dopamine	D_1 renal vasodilation
Asthma (acute)	Salbutamol	β_2 bronchodilation
Asthma (maintenance)	Salmeterol + ICS	Long β_2 bronchodilation
Hypertension (pregnancy)	Methyldopa	Central α_2 agonist
Nasal Congestion	Phenylephrine, Oxymetazoline	α_1 vasoconstriction
SVT	Phenylephrine, Methoxamine	Reflex bradycardia via α_1
ADHD / Narcolepsy	Amphetamine, Methylphenidate	$\uparrow$ CNS NE & DA release

References



- **Lippincott's** Illustrated Reviews: Pharmacology, 9th Edition. 2023
- **Katzung's** Basic & Clinical Pharmacology, 14th Edition. 2018
- **Rang and Dale's** Pharmacology, 8th Edition. 2021

